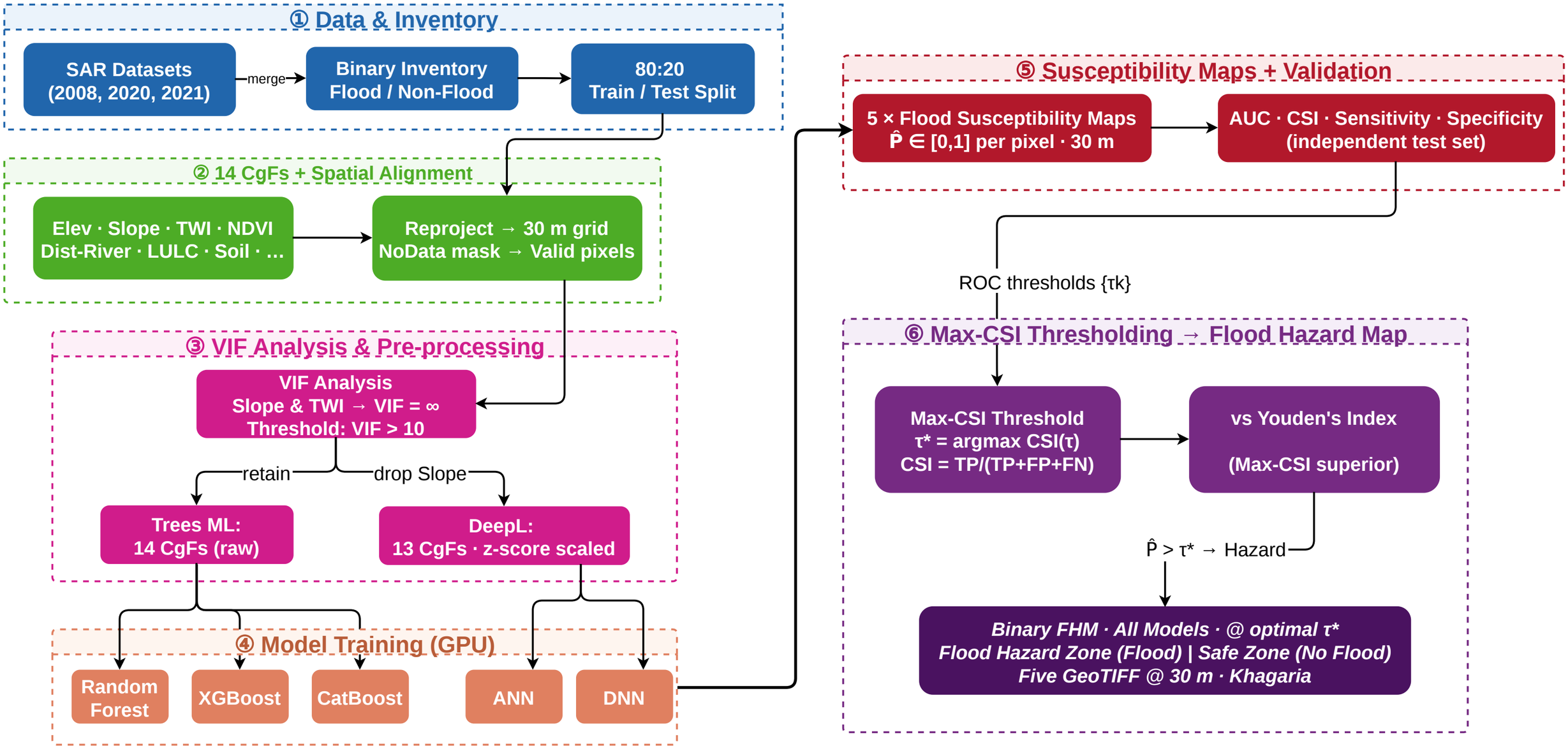

① Data & Inventory
SAR Datasets (2008, 2020, 2021)
merge
Binary Inventory Flood / Non-Flood
80:20 Train / Test Split
② 14 CgFs + Spatial Alignment
Elev · Slope · TWI · NDVI Dist-River · LULC · Soil · …
Reproject → 30 m grid NoData mask → Valid pixels
③ VIF Analysis & Pre-processing
VIF Analysis Slope & TWI → VIF = ∞ Threshold: VIF > 10
retain
drop Slope
Trees ML: 14 CgFs (raw)
DeepL: 13 CgFs · z-score scaled
④ Model Training (GPU)
Random Forest
XGBoost
CatBoost
ANN
DNN
⑤ Susceptibility Maps + Validation
5 × Flood Susceptibility Maps P̂ ∈ [0,1] per pixel · 30 m
AUC · CSI · Sensitivity · Specificity (independent test set)
ROC thresholds {τk}
⑥ Max-CSI Thresholding → Flood Hazard Map
Max-CSI Threshold τ* = argmax CSI(τ) CSI = TP/(TP+FP+FN)
vs Youden's Index
(Max-CSI superior)
P̂ > τ* → Hazard
Binary FHM · All Models · @ optimal τ*
Flood Hazard Zone (Flood) | Safe Zone (No Flood)
Five GeoTIFF @ 30 m · Khagaria